Artificial Superintelligence and Maxwell's Equations from a Pre-1855 Knowledge Base: Literature Review

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Query: Are there any references that describe artificial superintelligence successfully creating Maxwell's electromagnetic equations based on physics knowledge base prior to 1855?

Short Answer: No — No Direct Reference Exists

After an extensive search across academic papers, arXiv, vintage-LLM projects, and ASI thought-experiments, no work meets all four criteria simultaneously:

1. ASI-level system (not narrow symbolic regression)
2. Maxwell's complete equations, including the displacement current $\partial D/\partial t$
3. Knowledge cutoff < 1855, i.e. before Maxwell began 'On Faraday's Lines of Force' (1855-56)
4. Successful generative derivation, not just data-fitting

What does exist are partial analogues that fail on at least one criterion.

What a Pre-1855 Knowledge Base Actually Contains

By the first half of the 19th century the pieces were known but not unified. Before Maxwell's equations were postulated, several laws of electromagnetics were known: the Biot–Savart law, Ampere's circuital law, Gauss's law, Gauss's law for magnetism, and Faraday's law. These were for specific circumstances and mostly for static fields.

Historical timeline:

- 1780s: Charles-Augustin de Coulomb established his law of electrostatics
- 1825: André-Marie Ampère published his force law
- 1831: Michael Faraday discovered electromagnetic induction
- 1835: Carl Friedrich Gauss formulated flux law (not published until 1867)

Maxwell's own invention was the correction: Maxwell added displacement current to the electric current term in Ampere's Circuital Law. In his 1865 paper, A Dynamical Theory of the Electromagnetic Field, Maxwell used this amended version of Ampere's Circuital Law to derive the electromagnetic wave equation. Maxwell's brilliant correction was the introduction of the Displacement Current: $J_d = \varepsilon_0(\partial E/\partial t)$. That term is not directly observable in 1854 experiments. It was posited to fix charge-conservation inconsistency in Ampere's law — a theoretical leap, not an empirical curve-fit.

Academic Literature That Does Rediscover Maxwell, But Not Under Your Constraints

1. TheoSea (Stalzer & Ju, 2019) — Applied Sciences 9, 2899

The canonical reference for automated Maxwell rediscovery. The authors state: 'it may be within current computing and algorithmic technology to infer the Maxwell Equations directly from data given knowledge of vector calculus.' Abstract: 'There is sufficient information in the far-field of a radiating dipole antenna to rediscover the Maxwell Equations and the wave equations of light, including the speed of light c.' The paper also notes: 'Indeed, electrodynamics was the first unification (the electric and magnetic fields), and Einstein's special relativity is baked right into the equations.'

Why it does not satisfy your question:

• It uses a virtual experiment that already assumes Maxwell's equations to generate far-field E, B data — circular if you want a pre-1855 derivation.
• The language L includes modern vector calculus operators ($\nabla\cdot$, $\nabla\times$, $\partial/\partial t$) which were not formalized before 1855.
• It is a Julia meta-program performing compactness-driven enumeration, not an ASI, and runs in ~5 seconds on a laptop. No pre-1855 text or experimental corpus is used.

2. AI-Feynman (Udrescu & Tegmark, Science Advances 2020)

AI-Feynman is a physics-inspired method for symbolic regression. The authors report: 'We apply it to 100 equations from the Feynman Lectures on Physics, and it discovers all of them, while previous publicly available software cracks only 71.' This includes Coulomb, Gauss, and Faraday forms from Feynman Lectures, but as a symbolic regression benchmark with modern data, not a historical knowledge-base test. No displacement-current novelty claim.

3. AI-Newton (Peking University, 2025) and AI-Descartes

AI-Newton is explicitly framed as 'a concept-driven discovery system capable of autonomously deriving physical laws from raw data—without supervision or prior physical knowledge.' As a proof-of-concept it rediscovers Newton's second law, gravitation, and energy conservation from 46 Newtonian mechanics experiments. Electromagnetism is listed as future work; Maxwell is not demonstrated. AI-Descartes (IBM / Samsung / UMBC, Nature Communications 2023) combines symbolic regression with logical reasoning from background theory but reproduced Langmuir and Kepler laws, not Maxwell from pre-1855 axioms.

ASI Counterfactual / Vintage-LLM Experiments — Closest to Your Thought-Experiment

The idea you describe is exactly the Demis Hassabis AGI test: 'seal a model inside the knowledge of 1911, could it derive relativity by 1915?' This proposal spawned a whole family of vintage LLMs.

Definition: 'A vintage LLM is a language model trained (typically from scratch) on text from a bounded historical period, with no information from after a given knowledge-cutoff date in its training corpus.' These models enable counterfactual experiments (e.g. can a pre-1915 model independently arrive at general relativity?).

Implemented examples:
• TimeCapsuleLLM (1800-1875): Trained from scratch on London texts published between 1800 and 1875 — this is pre-1855 in principle, but used for social history (protests of 1834), not physics derivation.
• GPT-1900 / Machina Mirabilis (2026): 3.3B parameters built mostly on books and newspapers published before 1900. Mid-training included Maxwell's Treatise and Faraday's Experimental Researches, but with Einstein/relativity removed. It showed 'glimpses of intuition' — for example, occasionally asserting that light 'is made up of definite quantities of energy' or that gravity and acceleration are locally equivalent.
• Evaluation noted: 'An AI raised on pre-1900 books nearly rediscovered Einstein's light quanta' but with heavy caveats about reward hacking and failure on most tasks.

Crucially, no vintage LLM has been reported to derive Maxwell's full system from a pre-1855 cutoff. The GPT-1900 experiment deliberately included Maxwell's own Treatise (1865) in its physics mid-training, so it cannot test whether Maxwell could be created before Maxwell. A true pre-1855 experiment would need to exclude Maxwell entirely and would need to provide only Coulomb, Ampere, Biot-Savart, and Faraday data — and then ask the model to invent the displacement current.

Why This Specific ASI Demonstration Does Not Yet Exist

1. Definition of ASI: Bostrom-style ASI is defined as an intellect much smarter than best humans across every field, including scientific creativity. No current system qualifies; all systems above are narrow or LLM-based.

2. Epistemic leap: The hardest part of Maxwell is not fitting data but detecting the inconsistency in Ampere's law ($\nabla \cdot J \neq 0$ for time-varying charge) and postulating $\mu_0 \varepsilon_0 \partial E/\partial t$ to fix it. Current symbolic regression systems enumerate expressions that fit observed E(x,t), B(x,t); they do not perform consistency proofs.

3. Data circularity: If you generate training data from modern EM theory (dipole antenna far-field), you smuggle Maxwell in. A genuine pre-1855 corpus would contain only static and quasi-static measurements — insufficient to uniquely determine waves without the theoretical move Maxwell made.

What a Valid Future Reference Would Need to Show

To count as a positive reference for your question, a paper would need to document:
• Corpus: explicit list of sources ≤1854 (Coulomb torsion balance papers, Oersted 1820, Biot-Savart, Ampere's electrodynamic memoirs, Faraday's Experimental Researches series 1-29)

• Task: detect violation of continuity equation and propose corrected Ampere-Maxwell law plus prediction of EM waves with finite $c = 1/\sqrt{(\mu_0 \varepsilon_0)}$
• Metric: exact reproduction of the four modern equations, not just one of them
• No leakage: proof that Maxwell's 1861-65 papers, Heaviside vector formulation, and any post-1855 textbook derivations were excluded, including OCR footnotes.

No paper meeting this has been indexed as of October 2026.

References Consulted

• Stalzer & Ju (2019) Automated Rediscovery of the Maxwell Equations, Appl. Sci. 9, 2899
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